

Project Report

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Background:

Contemporary artificial intelligence tutoring systems rely predominantly on text documents and speech-to-text transcripts to generate educational assistance. Mainstream educational platforms such as Khanmigo and Coursera Coach, alongside document synthesis engines like Google NotebookLM, index linear transcripts while discarding visual video streams.

However, authentic technical and scientific instruction is fundamentally multimodal. Instructors frequently rely on deictic verbal references—such as "this cancellation yields the identity" or "the pointer dereference highlighted here"—while illustrating proofs on whiteboards, navigating architectural diagrams, or editing code in terminal windows. When educational AI assistants are blind to these visual artifacts, they produce severe conceptual hallucinations or offer vague, ungrounded generalities. Furthermore, existing consumer lecture summarizers rely on naive Optical Character Recognition (OCR) over periodic video screenshots, stripping essential spatial hierarchies, mathematical notations, and diagrammatic relationships.

To bridge this gap, an effective instructional AI must move beyond text-bound retrieval by ingesting synchronized video keyframes and audio tokens, producing verifiable sub-second timestamp citations, maintaining an auditable learner memory model, and enforcing calibrated abstention when grounding evidence is absent or deprecated.

Problem Statement:

Build an interactive AI tutor grounded exclusively in public course and video material. The system must retrieve information across text, video transcripts, slide images, and code snippets, moving beyond simple single-source text retrieval.

Key features to be implemented

- **Hybrid Multimodal Retrieval:** Combine sparse and dense retrieval (with optional fusion and reranking), evaluated across distinct query classes rather than through a single, blended metric.
- **Precision Deep-Linking:** Answers must include exact evidence references (such as direct video timestamps or specific visual assets) to allow seamless client navigation, rather than vague document-level citations.
- **Visual-Only Benchmark:** A designated subset of benchmark questions must rely solely on visual evidence (diagrams, slide graphics, or on-screen code) to prove the functional necessity of multimodal retrieval over raw transcript text.
- **Tiered Learner Memory:** Segment student context into two policy-governed layers rather than a single memory store:
 - Ephemeral State: In-session scratchpad data.
 - Durable State: Long-term goals, mastery levels, and logged misconceptions.
 - Governance: Explicit rules for write, expiry, correction, and deletion.
- **Abstention & Freshness Controls:** The system must actively abstain or qualify answers when queries fall outside the corpus. Additionally, it must detect, invalidate, and supersede stale citations when source material is updated.

Evaluation and Security

An automated evaluation harness must quantitatively benchmark the architecture across multiple dimensions:

- Retrieval quality, answer correctness, and citation support.
- Accurate abstention behaviour and stale-source invalidation.

- Resistance to indirect prompt injection (malicious instructions embedded in retrieved corpus content).
- Trade-off analysis measuring latency versus quality across at least two retrieval, chunking, and reranking configurations.

Key Challenges

- **Abstention vs. Multimodal Recall:** Strict abstention policies risk dropping valid answers if visual retrieval fails to surface slide or diagram evidence reliably.
- **Personalization vs. Security:** Expanding persistent learner memory improves instructional targeting, but it broadens the prompt-injection attack surface and increases the risk of storing hallucinated learner states.
- **Quality vs. Real-Time Latency:** Multi-stage hybrid search, fusion, and reranking improve response precision, but introduce latency costs that can degrade the live conversational flow.

Expected Output:

The target output is a fully functional, testable AI tutor system capable of multimodal reasoning across public course content, supported by an end-to-end evaluation harness.

The working prototype should include the following features:

- A versioned evidence corpus indexing text, transcript, slide/image/code evidence with timestamp and source/content-version metadata.
- A hybrid retrieval layer (sparse, dense, fused, optionally reranked) whose performance can be broken out by query class.
- An answer-generation layer that returns citations precise enough for a sample client/player to jump straight to the supporting video moment or asset.
- Can demonstrate correct performance on a benchmark subset that is unsolvable from transcript text alone, with the supporting visual evidence returned alongside the answer.
- A learner memory system that keeps ephemeral conversation state separate from durable mastery/misconception/goal data, with configurable write, expiry, correction, and deletion policies the learner or evaluator can exercise.

- Can demonstrate abstention/qualification on off-corpus or unsupported questions, plus detection and invalidation of stale evidence when source content changes.
- Can perform an automated evaluation and adversarial red-team harness reporting retrieval quality, correctness, citation support, abstention accuracy, stale-content handling, and resistance to malicious retrieved instructions.

The end deliverable is "done" when an evaluator can pose both text-only and visually-grounded questions over course material the system hasn't seen tuned for, inspect the exact evidence behind each answer, correct or delete stored learner memory, observe the tutor safely decline off-corpus questions, and independently reproduce the team's quality and adversarial benchmark results.